

Continuity



Testing continuity

- 1 Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.
 - a Find $\lim_{x \rightarrow 2^-} f(x)$.
 - b Find $\lim_{x \rightarrow 2^+} f(x)$.
 - c Is f continuous at $x = 2$? Justify.
 - 2 The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k .
-

Types of discontinuity and the Intermediate Value Theorem

- 3 Classify the discontinuity of each function at the given point:
 - a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$
 - b $f(x) = \frac{1}{x - 2}$ at $x = 2$
 - c $f(x) = \lfloor x \rfloor$ at $x = 3$
- 4 Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.